



Cracking Analytics Interview

Alumni Mentorship Programme
January 2021

1. Analytics 101 - 20 minutes
2. Sample Projects - 10 minutes
3. Career Growth and subsequent career options - 10 minutes
4. Learning hard skills - 5 mins
5. Solving Guesstimates - 30 minutes
6. Solving Case Studies - 30 minutes

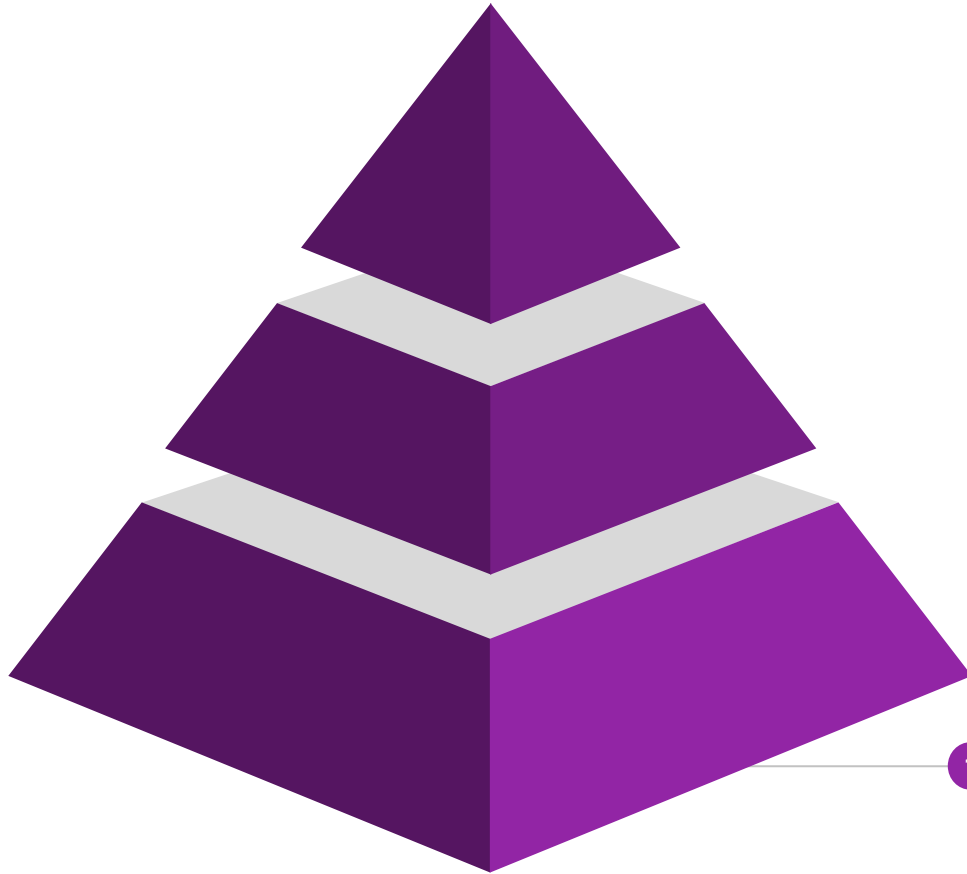
What are your favorite applications of data?

Here are some of mine

1. Cool

2. Crazy

3. Creepy

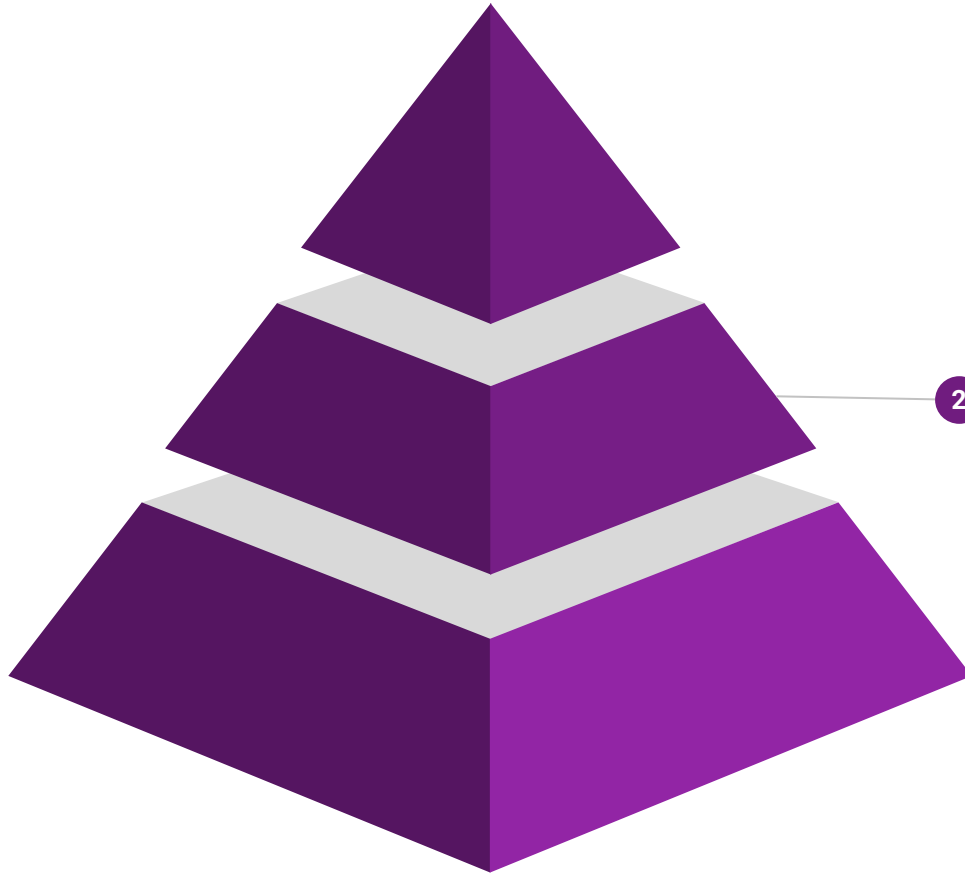


Data Engineer / Analyst / BI Developer

Work: ETL, maintaining and optimizing data pipelines and dashboards, RCAs

Skills: For DA, SQL and Python/R (good to have); Hive/NoSQL and Java/C (in case of ETL work) for DE; Tableau/PowerBI/Qlikview for dashboards

PayScale: Lowest



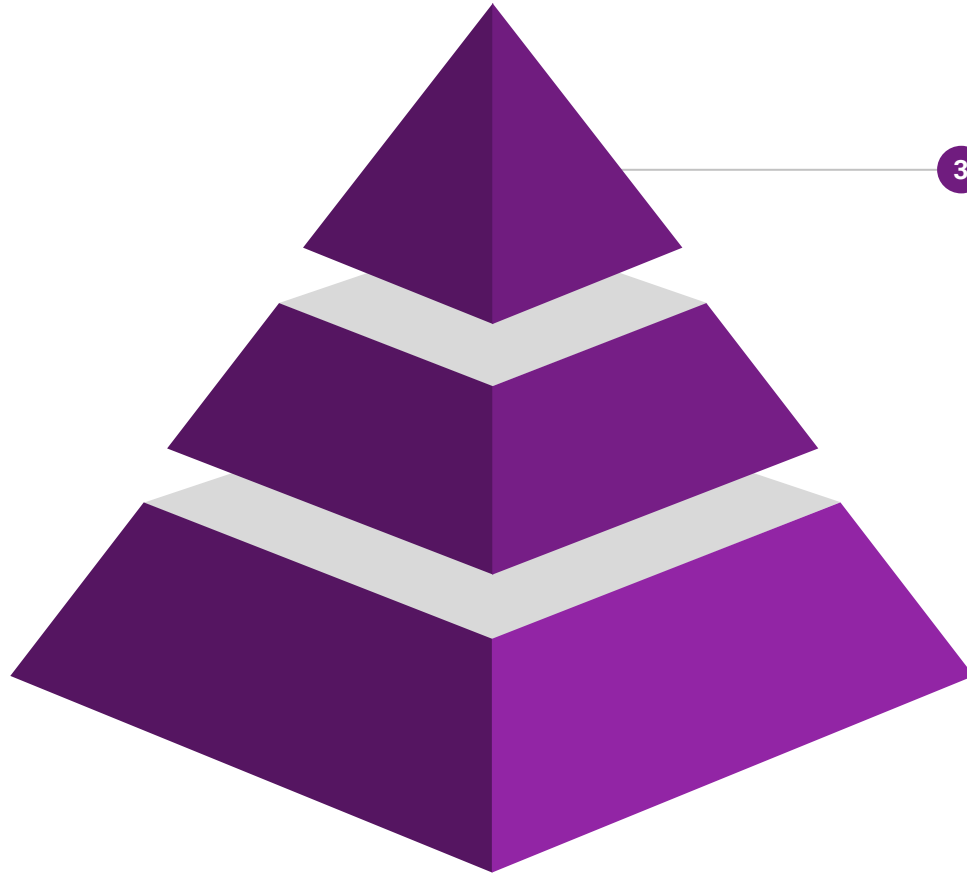
Business Analyst / Consultant

Work: Generating high value actionable insights from the data

2

Skills: SQL and Python/R (good to have), Basic Statistics, Excel, Powerpoint, Stakeholder Management, High level collaboration

PayScale: Mid-level



3

Data Scientist / Statistician

Work: Build, implement and scale extremely high value data projects using cutting edge techniques

Skills: SQL and Python/R, Advanced Statistics, Modelling, ML/AI, ability to understand and implement research papers, ability to work independently with very high sense of responsibility

PayScale: Highest

Some companies also offer **Machine Learning Engineer** role which involves implementing the solutions built by Analysts/Scientists on a software platform. It requires basic knowledge of both coding and Data Science

In most companies there is a **huge overlap** between these roles and responsibilities

Always **read Job description** before applying

And when HR calls, ask them about the work in detail

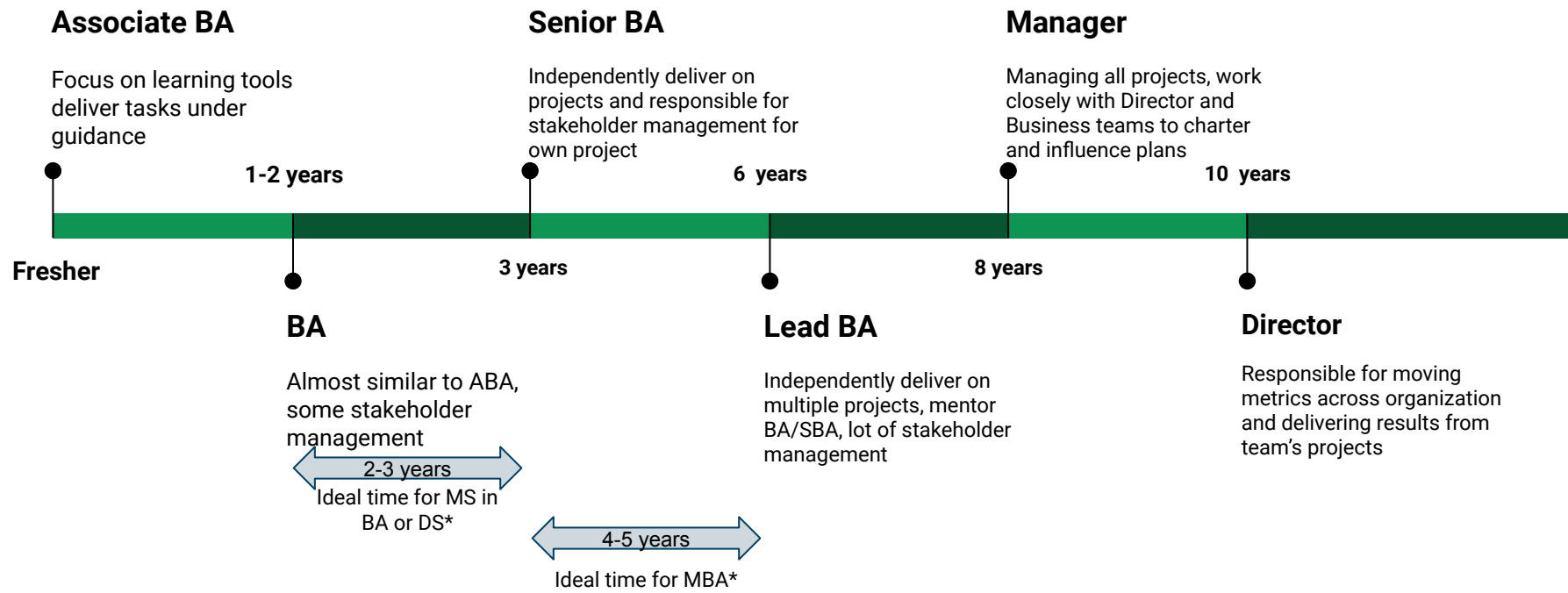
Business
Analyst

- e-Commerce - Right Price for a Product; Optimize Logistics; Product Placement on website
- Banking - Identify right customers for products; Optimize right risk constraints, Predict Customer LTV, Credit Scoring (e.g. CIBIL)
- Uber/Swiggy/Zomato - Correct value of surge pricing, Optimize driver/delivery person placement
- Pharma - Identify high risk customers, estimate efficacy of medicines
- Gaming - Optimize power levels of characters, multiplayer matchmaking
- All apps - Data powered UI/UX Designing e.g. via A/B testing
- Politics - Research on public issues, optimize ad spend

Data
Science

- Conversational AIs/Chatbots like Keya(Kotak Bank), Siri(Apple), Alexa (Amazon)
- Speech and image recognition like self driven vehicles(Tesla) and unmanned drones(DJI)
- Targeted Ads - Google/Facebook etc
- Anti-Money Laundering and Fraud Detection - all Banks have it!
- Recommendation Engines - Netflix, Amazon, Youtube etc.
- Content Ranking - Facebook Feeds, Google Search Results
- Gaming - Augmented Reality
- IOT - Synchronized smart devices
- Intelligent Robots to perform actions like Surgery, Warehouse Management, Boston Dynamics

Here is a sample career journey in Analytics



* timelines are indicative only and rather highly influenced by your career objectives

Where can you learn the required skills?

Must Know Skills

- **SQL**
 - Learn at [Codecademy](#) or Utsav's SQL for Data Analysis Course on [YouTube](#)
 - Practice at [Hackerrank](#) or [Sqlzoo](#)
- **Microsoft Excel**
 - Learn Vlookup, Pivot Tables and Basic Visualization on YouTube e.g. [Link 1](#), [Link 2](#)
- **Python or R** - Prefer learning Python
 - Learn Python at [Codecademy](#) and then learn pandas, numpy, matplotlib from Youtube e.g. [Link 1](#) OR Utsav's Python for Data Analysis Course on [YouTube](#)
 - Practice at [Hackerrank](#) or [PracticePython](#)
 - Once you are comfortable with basics, go through Kaggle Problems (HIGHLY RECOMMENDED)

Role Specific

- **Visualization** - Low RoI as different companies uses different visualization tools
 - Tableau or PowerBI or Qlikview from YouTube or their own websites
- **Basic Probability and Statistics**
 - Business Statistics and Analysis Specialization on [Coursera](#)
- **Machine Learning**
 - Intro to Statistical Learning from [Stanford](#) and then Machine Learning on [Coursera](#)
- **Deep Learning**
 - Deep Learning on [Coursera](#)

Practice real-world problems on Kaggle!

General Tips

- It's all about the approach - **logical thinking** and **structured communication** will take you places
- **ALWAYS** state your assumptions: think out loud
- Make assumptions that **simplify** your calculations
- Once you have arrived at an answer, **figure out if it makes sense**. You can use the demand side approach to check if you used the supply side to calculate, or just plain common sense.
- Keep it conversational, interviewer engaged
- What if there are **calculation errors**?
- **Structure the solution before** delving into numbers
 - Rural-Urban (Geography-wise)
 - Gender Split
 - Age Split
 - Income Split

Reference Material/Suggested books

- Vault Guide
- <http://www.streetofwalls.com/>

Numbers to remember

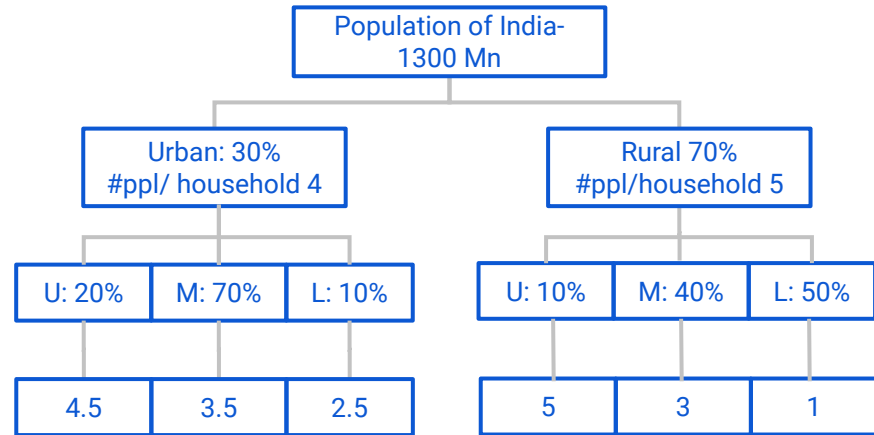
- Population = 1.3 billion ~ 1 billion
- Land Area = 3 million km²
- GDP growth rate = 5%
- Average size of a family = 3.6 ~ 4
- Population growth rate = 1.5 % (World = 1%)
- Sex ratio = 1:1
- Rural: Urban population = 70:30
- Upper Middle Class (>32,000 pm): 10%
- Middle Class (16,000-32,000 pm): 30%
- Lower Middle Class (8000-16,000 pm): 40%
- Below poverty line (<8000pm): 20%
- Kolkata, Delhi population (approx same for all metros) =15 million
- Population of your city, major metros

Different approaches

1. **Simple Math approach** - Used when the solve to the guesstimate is through some sort of a ratio based approach. Eg - number of tennis balls that can fill an airplane.
2. **Per capita approach** - Used when the number to guess can be thought of a consumption item at an individual, household or population level within a geography. Eg - number of smartphones in India
3. **Supply & Demand approach** - Used when we can constraint the case from either the production or the consumption side. Eg - Burgers sold by some McD outlet in a day.

Different approaches

1. Simple Math approach - Numerator/Denominator
2. What is the number of smartphones in India?



Urban households: $\rightarrow (1300 \text{ Mn} * .3 \text{ Urban})/4 = 97.5 \text{ Mn}$

Urban Upper class: $97.5 \text{ Mn households} * 20\% \text{ Upper class population} * 4.5 \text{ smartphones/household} = 87.75 \text{ Mn phones}$

< smartphones/household: $5*.5+4*.5 = 4.5 >$

Sample Guesstimate

- What is the revenue of a Pizza Hut outlet in a day?

Novice

- I think 100 people sit in an outlet at one time and I feel everyone pays 500 Rs per order.
- If we assume a rotation time of people to be 1 hr and if the store operates for 10 hrs in a day then
- Revenue = $100 \times 500 \times 10 / 1 =$
Rs 500000/ day

Advanced

- Ask clarification questions to understand the exact question being asked - **IMP**
Mention what is the rationale behind asking these questions - **initial hypothesis - IMP**
- We asked some of these -
 - Weekday? Weekend? Average?
 - What is the size of the outlet?
 - Where is the outlet?
 - Revenue only from Pizza?
 - Dine In only or Take out?
- Here, we come to a clear understanding of the question - What is the revenue from pizzas sold in a medium sized Pizza Hut Outlet in a metro during a Weekday for dine-in customers?

Sample Guesstimate

- What is the revenue from only the pizzas sold in a medium sized Pizza Hut Outlet in a Metro city during a Weekday for dine-in customers only? - Revised question you **only uncover when you ask questions**

Advanced

- Repeat the question in your words - make sure you and the interviewer are on the same page - both buys you some time and gives you clarity.
- Ask good clarifying questions at the outset itself to give you a clear understanding
- Structure your solution and then put numbers around your formula - helps reduce calculation errors. Use numbers/assumptions that make your calculations simpler.
- Make sure you state your assumptions clearly, and tell your interviewer as to why you took that assumption. Take their opinion on it, involve them, keep it conversational.
- Once you're done with the solution, try checking how logical the answer sounds.
- Synthesis - very important to finish strongly - state your answer upfront, then explain in a sentence as to how you came about it, and what the implications might be/any additional thoughts you have to better your solution if given the data.

Demand side approach can be used to calculate the revenue from pizzas.

Supply side to calculate take out revenue or to check if the demand side number and the answer using the Supply side approach converge. **Remember**: the final answer doesn't matter as long as the approach is logical.

Types of case questions

- Guesstimate (intention is to arrive at a number as the answer to the case problem)
- Strategy (multi-dimensional approach to solving the situation in hand)
- Process improvement (fix something that is broken or improve something that is not working as expected)

For analytics interviews - the above cases need to be solved from a “data” point of view

- What would be your approach
- What data would you need/use and from what source
- What would be your hypothesis
- What would be those assumptions that need to be true to arrive at a trustworthy answer

Common frameworks - pick one, stick to that

- Profit equation
- Pricing methods (Ex. value based, cost plus, parity)
- 4Ps framework (Product, Price, Place, Promotion)
- 3C framework (Customer, Company, Competitor)
- PESTEL analysis

Sample questions

- Profit of company XYZ reducing every month. Why?
- Customers purchasing goods from XYZ can return the products within 14 days of purchase. Suppose they sell 1000 SKUs per day in a city
 - How many trips for collecting returns should they account for?
 - How to reduce return rate without reducing NPS?
- How to grow the books category of company XYZ
- Where should an established ecom company focus to maximize growth - Tier 1, 2 or 3 cities?

Profit of company XYZ reducing every month. Why?

Profit = Revenue - Cost

- Have the sales reduced?
- Overall sales or sales per person?
 - People buying product less often (buying rice every month vs every fortnight)
 - People buying lesser quantity (buying 1kg rice every time vs 2kg rice)
- What has changed?
 - Product (features, functionality)
 - Price (real, perception)
 - Place (distribution channels)
 - Promotion (marketing)
- What has changed?
 - Customer (preference, use)
 - Competitor (their 4P)
 - Company

Profit = Revenue - Cost

- Has the cost increased?
 - Fixed cost (overheads, licenses, fees)
 - Variable cost (per product)
 - Raw material
 - Labour
 - Utilities
 - Why has the cost increased?
 - External (price, logistics, taxes)
 - Internal (processing time has increased - more layers of QA)

Good Business
Exposure while
working with
multiple teams

Competitive Salary

Easy to justify
career switch for
MBA

Exciting problems
to Solve

Can influence key
Business
Decisions

Intellectually
stimulating, teaches
story-telling and
improves critical
thinking

Travel perks in
many companies

Many companies
are offering flexible
working hours and
even WFH

Good Data
Scientists are
highly paid and low
in supply

Thank You!

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